

Exercise 4

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Question 1-5 Extract Date Parts

① Day name from order date

```
SELECT order-id, customer-id, order-date,  
       DATENAME(order-date) AS day-name  
FROM Orders;
```

Order-id	Cust-id	Order-date	day-name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

② Month name from signup date

```
SELECT cust-id, cust-name, signup-date, signup-Mon-Name  
       MONTHNAME(signup-date) AS Month-name  
FROM Customer-Signups;
```

Cust-id	Cust-name	signup-date	signup-month-name
201	John	2026-01-15	January
202	Mary	2026-02-20	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

③ Month number from sale date

```
SELECT sale-id, product-name, sale-date, Sale-Month
      MONTH(sale-date) AS Sale-month
FROM sales;
```

Sale-id	Product-name	Sale-date	Sale-month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

④ Year from transaction date

```
SELECT trans-id, cust-id, trans-date, trans-year
      YEAR(trans-date) AS transaction-year
FROM transactions;
```

trans-id	Cust-id	transaction-date	transaction-year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

⑤ Day of Month from delivery date

```
SELECT delivery-id, cust-id, delivery-date, delivery-year
      DAY(delivery-date) AS day-of-Month
FROM deliveries;
```

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delivery-id	cust-id	delivery-date	day-of-month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

Question 6-10 Current date, Convert & Format

(6) Add today's date

SELECT employee-id, employee-name, department, today-name
CURRENT-DATE() AS today-date

FROM employees;

Employee-id	employee-name	department	today-date
301	Nandi	Sales	2026-07-19
302	Brian	IT	2026-07-19
303	Lerato	Finance	2026-07-19
304	Sipho	HR	2026-07-19
305	Aisha	Marketing	2026-07-19

(7) Convert text to date

SELECT order-id, cust-id, order-date-text, Order-date
TO_DATE(order-date-text, 'YYYY-MM-DD') AS Order-date

FROM Online-orders;

Order-id	cust-id	order-date-text	order-date
4001	101	2026-01-15	2026-01-15
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-04-10	2026-04-10
4005	105	2026-05-05	2026-05-05

⑧ format date as text

SELECT payment-id, cust-id, payment-date, formatted-pay-date
 TO_CHAR(payment-date, 'YYYY-MM-DD') AS formatted-pay-date
 FROM payment-dates;

Payment-id	cust-id	payment-date	formatted-pay-date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-25

⑨ Day since last purchase

SELECT cust-id, cust-name, last-purchase-date, day-since-last-purchase
 DATEDIFF(CURRENT-DATE(), last-purchase-date) AS day-since-last-purchase
 FROM Customer-purchases;

Cust-id	cust-name	last-purchase-date	day-since-last-purch
501	John	2026-05-01	79
502	Mary	2026-05-10	70
503	Peter	2026-05-15	65
504	Sarah	2026-05-20	60
505	Thabo	2026-05-25	55

⑩ Expected delivery date

```
SELECT order-id, cust-id, order-date, expected-delivery-date
DATEADD(DAY, 7, order-date) AS expected-delivery-date
FROM shipping-orders;
```

Order-id	cust-id	Order-date	expected-delivery-date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

Questions 11-15 multi-extract & date filtering

⑪ Year, month, day separately

```
SELECT booking-id, customer-id, booking-date
YEAR(booking-date) AS booking-year
MONTH(booking-date) AS booking-month
DAY(booking-date) AS booking-day
FROM bookings;
```

booking-id	cust-id	booking-date	booking-year	booking-month	booking-day
B001	101	2026-01-12	2026	01	12
B002	102	2026-02-18	2026	02	18
B003	103	2026-03-22	2026	03	22
B004	104	2026-04-09	2026	04	09
B005	105	2026-05-27	2026	05	27

(12) Filter orders from 2026

SELECT order-id, cust-id, order-date,
 YEAR(order-date) AS order-year, amount
 FROM yearly-orders
 WHERE YEAR(order-date) = 2026;

Order-id	Cust-id	Order-date	Order-year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

(13) Filter orders from March

SELECT order-id, cust-id, order-date
 MONTH(order-date) AS order-month, amount
 FROM monthly-orders
 WHERE MONTH(order-date) = 3;

Order-id	Cust-id	Order-date	Order-month	amount
8003	103	2026-03-10	3	800
8004	104	2026-03-25	3	1500

(14) Last day of the month

SELECT Subscription-id, Customer-id, start-date
 LAST_DAY(start-date) AS month-end-date
 FROM Subscriptions;

Subs_id	Cust_id	Start_date	Month_end_date
SUB001	101	2026-01-10	2026-01-31
SUB002	102	2026-02-15	2026-02-28
SUB003	103	2026-03-20	2026-03-30
SUB004	104	2026-04-25	2026-04-30
SUB005	105	2026-05-30	2026-05-31

(15) First day of the Month

```
SELECT Subs_id, Customer_id, Send_date
       DATE_TRUNC(Month, Send_date) AS Month_start_date
FROM Campaign_Sends;
```

Send_id	Cust_id	Send_date	Month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

Questions 16-20 advanced date work

(16) Format as Month and year

```
SELECT invoice_id, Cust_id, invoice_date,
       TO_CHAR(invoice_date, 'Month YYYY') AS Inv_Month_year
FROM invoice_dates;
```

Invoice-id	Cust-id	invoice-date	Invoice-month-year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

(17) Customer Age from date of birth

```
SELECT cust-id, cust-name, date-of-birth,
DATEDIFF(CURRENT_DATE(), date-of-birth) / 365, 25) AS
cust-age
FROM customer-birthdays;
```

Cust-id	customer-name	date-of-birth	Cust-age
901	John	1998-05-10	28
902	Mary	1990-08-20	35
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	40
905	Thabo	2000-07-30	25

(18) Classify Weekend vs Weekday

```
SELECT order-id, customer-id, order-date,
DAYNAME(order-date) AS day-name,
CASE
WHEN DAYNAME(order-date) IN ('Saturday', 'Sunday') THEN
Weekend
ELSE 'Weekday'
END AS day-type
FROM weekend-orders;
```

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Order-id	Cust-id	Order-date	day-name	day-type
9001	101	2026-05-01	Friday	Weekdy
9002	102	2026-05-02	Saturday	Weekend
9003	103	2026-05-03	Sunday	Weekend
9004	104	2026-05-04	Monday	Weekdy
9005	105	2026-05-05	Tuesday	Weekdy

(19) Transactions by Quarter

```
SELECT trans-id, cust-id, trans-date
      Quarter(trans-date) AS trans-quarter, amount
FROM quarterly-transactions;
```

trans-id	Cust-id	trans-date	trans-quarter	Amount
Q001	101	2026-01-15	1	500
Q002	102	2026-03-20	1	1200
Q003	103	2026-04-10	2	800
Q004	104	2026-07-05	3	1500
Q005	105	2026-10-25	4	2000

(20) Orders older than 30 days

```
SELECT order-id, cust-id, order-date
      DATEDIFF(CURRENT_DATE(), order-date) AS days-since-order
      amount
FROM recent-orders
WHERE DATEDIFF(CURRENT_DATE, order-date) > 30;
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Order-id	Cust-id	Order-date	days-Since-Order	amount
R001	101	2026-04-01	109	500
R002	102	2026-04-15	95	1200
R003	103	2026-05-01	79	800
R004	104	2026-05-10	70	1500
R005	105	2026-05-25	55	2000